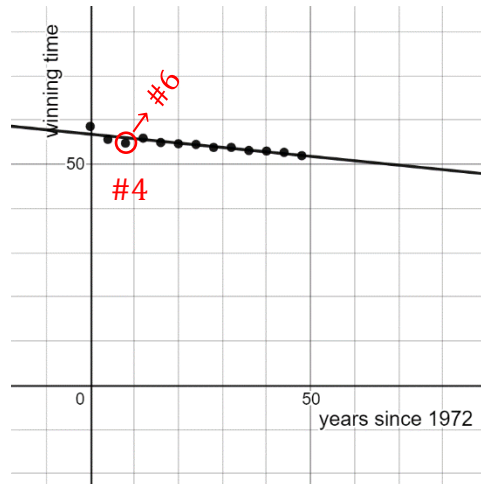


A scatter plot representing the gold medal winning times for the womens' 100-meter freestyle swimming race since 1972 is shown.



The raw data used to construct the scatter plot is shown.

Year	1972	1976	1980	1984	1988	1992	1996
Winning Time (in seconds)	58.59	55.65	54.79	55.92	54.93	54.65	54.5
Year	2000	2004	2008	2012	2016	2020	
Winning Time (in seconds)	53.83	53.84	53.12	53	52.7	51.96	

The equation of the line of best fit that models this data is $f(x) = -0.1x + 56.83$, where f is the winning time of the womens' 100 meter freestyle even, and x is the number of years since 1972.

- How many years after 1972 was the 2016 Olympics?

44 years

- Using the equation of the line of best fit, find the predicted finishing time of the gold medalist in 2016.

$$f(x) = -0.1(44) + 56.83 \quad f(x) = 52.43$$

The predicted finishing time is 52.43 seconds.

- Identify the residual value of the point representing the 2016 winning time.

$$52.7 - 52.43 = 0.27$$

4. Identify one point on the scatter plot where the predicted value is above the actual value.

1980 Predicted: 56.03
Actual : 54.79

5. Find the residual value of the point selected for question 4.

$$54.79 - 56.03 = -1.24$$

6. Identify one point on the scatter plot where the predicted value is below the actual value.

1984 Predicted: 55.63
Actual : 55.92

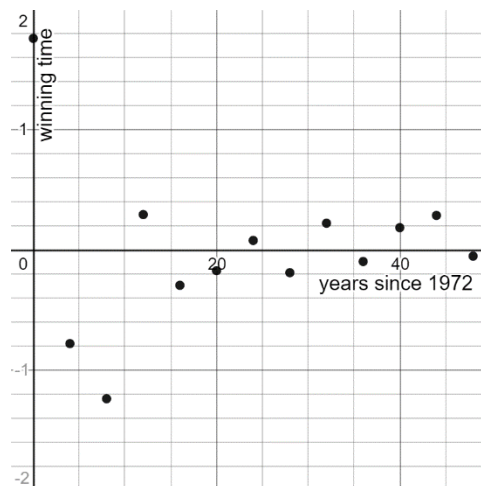
7. Find the residual value of the point selected for question 6.

$$55.92 - 55.63 = 0.29$$

8. Does the data appear to have a strong linear correlation? Explain how you know.

Because the data fits so closely with the line of fit, it is a strong linear correlation.

The residual plot for the scatter plot is shown.



9. Explain how the residual values support the answer from question 8.

Because so many residuals are less than 1 & greater than -1 on the y-axis, this supports the idea that because the residuals are so small it is a stronger linear correlation.